

Received 20 October 2025, accepted 5 November 2025, date of publication 11 November 2025,
date of current version 20 November 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3631711

RESEARCH ARTICLE

Performance Analysis and Optimization of Terahertz Metamaterial Absorbers Using Machine Learning-Based Inverse Modeling

OISHI JYOTI¹, (Member, IEEE), MD. SAMIUL HABIB^{1b2}, (Senior Member, IEEE),
NGUYEN HOANG HAI^{1b3}, (Member, IEEE), AND
S. M. ABDUR RAZZAK², (Senior Member, IEEE)

¹Department of Electrical & Computer Engineering, Rajshahi University of Engineering & Technology, Rajshahi 6204, Bangladesh

²Department of Electrical & Electronic Engineering, Rajshahi University of Engineering & Technology, Rajshahi 6204, Bangladesh

³School of Electrical and Electronics Engineering, Hanoi University of Science and Technology, Hanoi 100000, Vietnam

Corresponding authors: Md. Samiul Habib (samiul.habib@ruet.ac.bd) and Nguyen Hoang Hai (hai.nguyenhoang@hust.edu.vn)

ABSTRACT We present a tunable broadband terahertz (THz) metamaterial absorber with a structurally simple, single-layered vanadium dioxide (VO₂) elliptical ring resonator. This design achieves a wide, near-perfect absorption band (3.5–5 THz) without the need for complex multi-layer stacks or hybrid-patterned alternatives. Full-wave simulations demonstrate that VO₂'s insulator-to-metal transition dynamically enhances absorption, while structural parameters—ring width, ellipticity ratio, and dielectric thickness—precisely control bandwidth and spectral response, as explained by impedance matching theory and electric field distributions. Furthermore, we explore the impact of varying the angle of incidence, highlighting the angular sensitivity of the structure. **Beyond conventional parametric sweeps, we implement a targeted machine learning (ML) strategy for inverse design.** Our models, trained on augmented data, show that **Random Forest Regressor** excels in predicting multiple geometric parameters simultaneously, while **CatBoost** is optimal for single-target prediction. **The predicted geometric parameters are validated through simulation; this ML-guided approach,** tailored to different design goals, combines physics-based modeling with data-driven optimization, offering a robust and efficient framework for designing next-generation broadband THz absorbers.

INDEX TERMS Metamaterial, absorber, vanadium dioxide.

I. INTRODUCTION

The terahertz (THz) regime spanning from 0.1 THz to 10 THz has emerged as a promising spectral range for a range of applications such as biomedical imaging [1], security [2], wireless communication [3], sensors [4], modulators [5] and absorbers [6], however, developing efficient, compact, and tunable THz absorbers remain a challenge due to limitations in conventional materials and fabrication techniques. Metamaterial absorbers (MMAs) have emerged as a transformative solution for THz wave manipulation through engineered subwavelength structures, enabling tailored electromagnetic responses such as impedance matching and resonant wave

trapping to achieve near-perfect absorption at target frequencies [7]. Most existing metamaterial based electromagnetic absorbers typically employ a tri-layer architecture consisting of a patterned resonator, a metallic ground plane, and an intermediate dielectric spacer, which can be optimized for narrow-band [7], multi-band [8], [9], or broad-band [10], [11], [12], [13] absorption depending on application requirements. While these absorbers generally exhibit favorable characteristics such as wide-angle absorption and moderate tunability, their performance is fundamentally constrained by plasmonic resonance mechanisms that produce fixed, narrow-band absorption peaks—a limitation first demonstrated experimentally in 2008 with an ultra-thin microwave absorber [7]. This inherent trade-off between structural compactness and operational bandwidth has driven ongoing

The associate editor coordinating the review of this manuscript and approving it for publication was Sukhdev Roy.

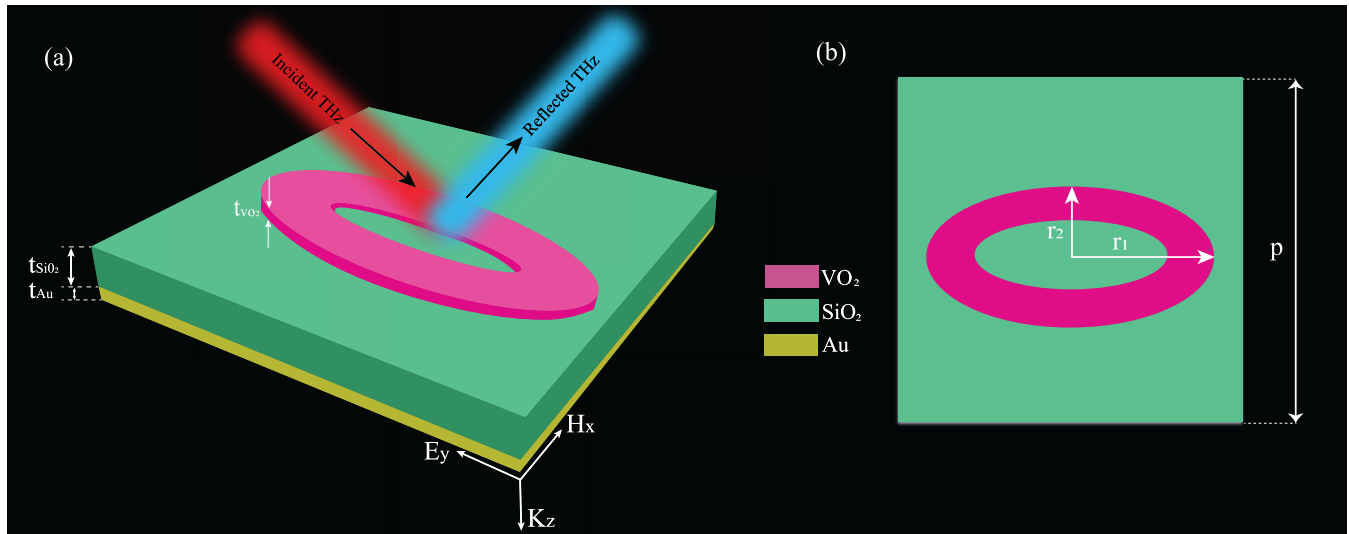


FIGURE 1. (a) A 3D schematic of the proposed metamaterial absorber, depicting the orthogonal field components and propagation vector: the magnetic field (H_x) along the x-axis, electric field (E_y) along the y-axis, and wave vector (K_z) along the z-axis. In Fig. 1(a), t_{SiO_2} , t_{Au} , and t_{VO_2} denote the thicknesses of silicon-dioxide, gold, and vanadium-dioxide layers, respectively. (b) Top view of the unit cell. In Fig. 1(b), p denote the period, and r_1 and r_2 represent radius of the VO₂ ring along the semi-major axis and semi-minor axis, respectively.

research into alternative designs capable of combining subwavelength thickness with broadband or dynamically tunable absorption capabilities in the THz regime. To address the inherent limitations of conventional MMAs and expand the versatility of them, reconfigurable designs have been developed using tunable materials, such as graphene [6], liquid crystals [14], semiconductors [15], and phase-change materials [16]. Among them, phase-change materials such as vanadium dioxide (VO₂) have been extensively investigated due to their reversible insulator-to-metal transition near 68°C, which induces orders-of-magnitude changes in electrical conductivity and enables dynamic THz absorption tuning [8], [17]. Note that the phase-change properties of VO₂ are sensitive to temperature. The existing literature extensively documents VO₂-based metamaterial in which tunable electromagnetic properties are achieved through thermal modulation [17], [18]. For example, Xu *et. al.*, first proposed the integration of a fractal MEMS micro-heater beneath an H-shaped VO₂ absorber [19]. In their approach, applying a precise DC bias to the micro-heater produces uniform thermal energy, thus directly modulating the electrical conductivity of the VO₂ layer. This mechanism enables active control over absorption characteristics. Moreover, VO₂-based metamaterial absorbers have been successfully demonstrated through experimental studies, exhibiting dynamic performance across a broad spectrum from the THz to the infrared frequency bands [20], [21]. This wide operational range highlights their significant potential for applications in tunable photonics and reconfigurable metamaterial devices. Recent studies have demonstrated that VO₂-based absorbers offer more than 90% absorption with either broadband or multi-band characteristics, although often at the cost of increased structural complexity [16], [17], [22]. For example, THz absorbers based on VO₂ exhibit diverse design approaches with

distinct trade-offs: while crossed-resonator configurations offer strong tunability but face fabrication difficulties, periodic VO₂ structures achieve enhanced bandwidth (0.65 THz) through multiple resonances, albeit with increased geometric complexity [23]. Thin VO₂ metasurfaces show particularly strong performance, with simulations demonstrating absorption tunability from 5% to near-perfect levels across 0.47–0.99 THz via conductivity modulation [24]. More advanced double-layer designs push performance further, enabling ultra-broadband operation (1.63–12.39 THz, 153.5% relative bandwidth) and multi-band functionality useful for a range of applications [25]. However, these improvements typically require complex multilayer structures or heterogeneous unit cells, making practical implementation challenging and highlighting the critical need for absorber designs that optimally balance tunability, bandwidth, and manufacturability. Moreover, multi-band absorbers show significant potential for diverse applications including sensors [26], satellite communications [27], microstrip antennas [28], and solar cells [29], but their practical implementation is often hindered by complex multilayered structures or intricate geometrical designs that pose fabrication challenges and complicate integration into functional devices. Moreover, significant advancements are continually being made in materials and devices central to opto-electronics and semiconductor applications. This progress is largely driven by the integration of new substances such as aluminum nitride (AlN) [30], MXene compound (Ti_2CO_2) [31], Janus monolayers including WSSe, MoSSe [32] and the silicon-carbon cluster (Si_9C_{15}) [32]. Researchers are harnessing the unique properties of these materials to develop highly sensitive photodetectors, substantially enhance charge carrier mobility for faster electronics, and fundamentally improve the physical and chemical characteristics of components [13],

[33], [34]. These enhancements are critical for driving innovation in next-generation energy harvesting systems and high-performance optoelectronic devices. Furthermore, the emergence of advanced two-dimensional and heterojunction materials represents a paradigm shift in optoelectronics, offering a unique combination of superior optical absorption, high carrier mobility, and precisely tunable electronic properties. These materials thereby establish a new materials platform, paving the way for a new generation of high-performance photovoltaic, photodetector, and terahertz absorber technologies [30], [35], [36].

Recent advances in functional materials have driven the adoption of machine learning (ML) for optimizing metamaterial absorber designs, enabling rapid performance prediction and streamlined optimization. While VO₂-based absorbers have emerged as a prominent platform due to their exceptional tunability, ML techniques further enhance their design efficiency and accuracy. However, despite the high absorption performance of many contemporary designs, their structural complexity often limits practical implementation. Traditional design approaches relying on iterative parameter adjustments are prohibitively time-consuming, creating a pressing need for simpler, manufacturable absorbers that maintain high performance across target frequency bands. ML addresses these challenges by offering data-driven solutions—from predicting broadband absorption spectra for complex solar absorber geometries to enabling computationally efficient multi-band optimization [37], [38], [39]. Notably, ML has demonstrated remarkable capabilities in navigating high-dimensional design spaces, as demonstrated in the inverse design of THz absorbers [40], while neural networks have facilitated real-time parameter tuning for tunable-based systems to meet precise absorption requirements [41].

In this work, we introduce a structurally simplified VO₂ based metamaterial absorber, utilizing a single-layer elliptical ring resonator to achieve broad, tunable bandwidth—overcoming the need for the complex multilayer structures or hybrid structures. This design offers enhanced flexibility and easier fabrication. Through full-wave numerical simulations, we analyze the absorber's performance with varying geometric and material conditions. To further optimize the design, we employ machine learning models trained on simulation data, enabling efficient prediction of structural parameters for target absorption spectra. By integrating physics-based simulations with data-driven ML techniques, our approach establishes a robust framework for developing tunable, broadband THz absorbers with high manufacturability. Furthermore, we demonstrate the design's robustness to simulated manufacturing errors, underscoring its high manufacturability and paving a credible path towards real-world applications in sensing and imaging.

II. DESIGN AND SIMULATION

We present a tunable THz absorber based on a VO₂ elliptical ring resonator, with its 3D geometry illustrated in Fig. 1(a), and the top view of the unit cell is depicted in Fig. 1(b). The

design consists of a VO₂ elliptical ring having a thickness of 5 μm atop a SiO₂ dielectric spacer and a bottom metal ground plane, arranged in a periodic array with periodicity of 30 μm . The elliptical resonator features a semi-major axis ($r_1 = 12 \mu\text{m}$) and semi-minor axis ($r_2 = 7.5 \mu\text{m}$), maintaining an aspect ratio, η of 1.6 to optimize both absorption peak and operational bandwidth. The structure's bottom layer consists of gold, serving as the ground layer. This gold layer has a thickness of 0.2 μm and exhibits a high electrical conductivity of $4.56 \times 10^7 \text{ S/m}$ [8]. The elliptical VO₂ ring has a ring radius of 12 μm and 7.5 μm along the semi-major axis and semi-minor axis, respectively. A SiO₂ spacer layer, 9 μm thick with a relative dielectric constant (ϵ) of 3.8, separates the gold layer from the VO₂ ring. This single-layer elliptical geometry provides enhanced control over resonant characteristics compared to circular designs, while the VO₂ phase transition enables dynamic tunability. Through systematic variation of ellipticity ratio (η) we investigate how geometric parameters influence the absorber's electromagnetic response, particularly focusing on field confinement and impedance matching mechanisms. The optimized structural parameters of the proposed absorber, determined through numerical simulations, are provided in Table 1.

In the terahertz regime, VO₂ exhibits a dominant metallic character that is well-captured by the Drude model, providing critical insights into electron transport phenomena, directly linking the material's frequency-dependent complex dielectric function, $\epsilon(\omega)$, which can be expressed as [8]:

$$\epsilon(\omega) = \epsilon_\infty - \frac{\omega_p^2}{(\omega^2 + i\gamma\omega)}, \quad (1)$$

where, $\epsilon_\infty = 12$ is the high-frequency dielectric constant and $\gamma = 5.75 \times 10^{13} \text{ rad/s}$ corresponds to the electron collision frequency [8]. The relationship between plasma frequency (ω_p) and conductivity (σ) can be described as [8]:

$$\omega_p^2(\sigma) = \frac{\sigma}{\sigma_o} \omega_p^2(\sigma_o), \quad (2)$$

where $\sigma_o = 3 \times 10^{15} \text{ S/m}$ and $\omega_p(\sigma_o) = 1.4 \times 10^{15} \text{ rad/s}$. This equation demonstrates how variations in electrical conductivity directly modulate the material's interaction with incident electromagnetic radiation. Note that the thermally induced phase transition drives VO₂'s conductivity from 200 S/m in the insulating phase to $2 \times 10^5 \text{ S/m}$ in the metallic phase [16].

In this work, we implemented the proposed structure and conducted full-wave simulations in CST Studio Suite using a tetragonal mesh to optimize computational efficiency while maintaining the accuracy to a great extent. The structure was excited by an electromagnetic wave in the fundamental TE₀₀ mode, with unit cell boundary conditions applied in the transverse (x - y) plane and open boundaries along the propagation direction (z -axis), replicating realistic wave propagation in the structure. The frequency-dependent absorption of the structure, $A(\omega)$, is calculated using the

TABLE 1. Optimized parameter list of the proposed absorber.

Parameter	P	t_{Au}	t_{SiO_2}	t_{VO_2}	r_1	r_2	η
Value	$30 \mu\text{m}$	$0.2 \mu\text{m}$	$9 \mu\text{m}$	$0.2 \mu\text{m}$	$12 \mu\text{m}$	$7.5 \mu\text{m}$	1.6

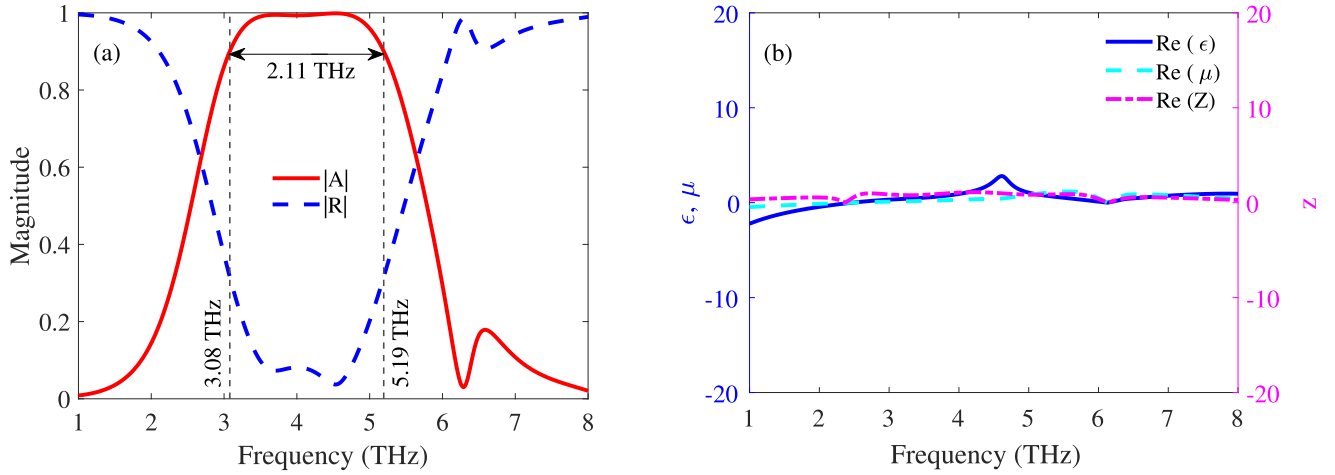


FIGURE 2. (a) Calculated frequency-dependent absorption (red) and reflection (dotted blue) spectra of the absorber, computed for a conductivity, $\sigma = 2 \times 10^5 \text{ S/m}$. **(b)** Frequency-dependent real part of the relative permittivity, $\Re(\epsilon)$, permeability, $\Re(\mu)$ (left axis) and real part of relative impedance, $\Re(Z_r)$ (right axis) of the absorber, extracted from the complex S-parameters.

following formula [16]:

$$A(\omega) = 1 - R(\omega) - T(\omega) = 1 - |S_{11}(\omega)|^2 - |S_{21}(\omega)|^2, \quad (3)$$

Here, $R(\omega) = |S_{11}|^2$ represents the reflection coefficient, and $T(\omega) = |S_{21}|^2$ represents the transmission coefficient. This means that the absorption is determined by subtracting the reflected and transmitted power from the total incident power.

III. RESULTS AND DISCUSSION

Our study focuses on developing a high-performance broadband absorber with exceptional absorption characteristics. As shown in Fig. 2(a), when the VO_2 layer exhibits metallic properties with a conductivity of $2 \times 10^5 \text{ S/m}$ and a $5 \mu\text{m}$ ring width, the absorber shows remarkable performance—achieving near-unity absorption in the 3.5–5 THz range and maintains over 90% absorption across an ultra-wide 2.11 THz bandwidth ranging from 3.08 THz to 5.19 THz. The relative absorption bandwidth, a crucial parameter for measuring the broadband absorption performance, can be expressed as [42]:

$$RAB = \frac{2 \times (f_H - f_L)}{f_H + f_L}, \quad (4)$$

In which f_H and f_L are, respectively, defined as the highest and lowest frequencies where the absorption rate exceeds 90%. The proposed metamaterial absorber achieves an impressive 51.03% RAB, as calculated by Eq. 4, underscoring exceptional broadband performance. These results represent significant improvements over previous designs, both in terms of peak absorption efficiency and operational

bandwidth. Figure 2(b) displays a dual-axis plot comparing the structure's impedance (right y-axis) with its frequency-dependent permittivity, ϵ and permeability, μ (left y-axis). At VO_2 's metallic conductivity ($2 \times 10^5 \text{ S/m}$), the real part of the impedance approaches unity while the imaginary component vanishes at 4.5 THz—precisely matching the absorption peak frequency. This impedance convergence with free space, satisfying the fundamental criteria for perfect absorption. Concurrently, the near-equality of the real part of permittivity and permeability values further confirms the structure's optimized impedance matching, explaining its exceptionally low reflectivity and high absorption efficiency at the design frequency.

Figure 3(a) systematically examines how the dielectric layer thickness, t_{SiO_2} varied from $1 \mu\text{m}$ to $9 \mu\text{m}$ affects absorption performance. The results demonstrate a clear thickness-dependent trend: while the thinnest configuration $1 \mu\text{m}$ yields minimal absorption (≈ 0.05), increasing the thickness progressively enhances performance, ultimately achieving perfect absorption ($\approx 100\%$) at $9 \mu\text{m}$ across a 1.5 THz bandwidth. This pronounced improvement, spanning two orders of magnitude in absorptance, underscores the critical role of dielectric thickness in optimizing the absorber's electromagnetic response. Based on these findings, we selected the $9 \mu\text{m}$ thickness for subsequent analysis as it represents the optimal configuration for maximum broadband absorption. Figure 3(b) analyzes the absorption spectra across 1–8 THz for varying metal layer thicknesses, t_{Au} ranging from $0.1 \mu\text{m}$ to $0.9 \mu\text{m}$, revealing a distinct thickness-dependent performance trend. The results demonstrate a non-monotonic relationship, where absorption

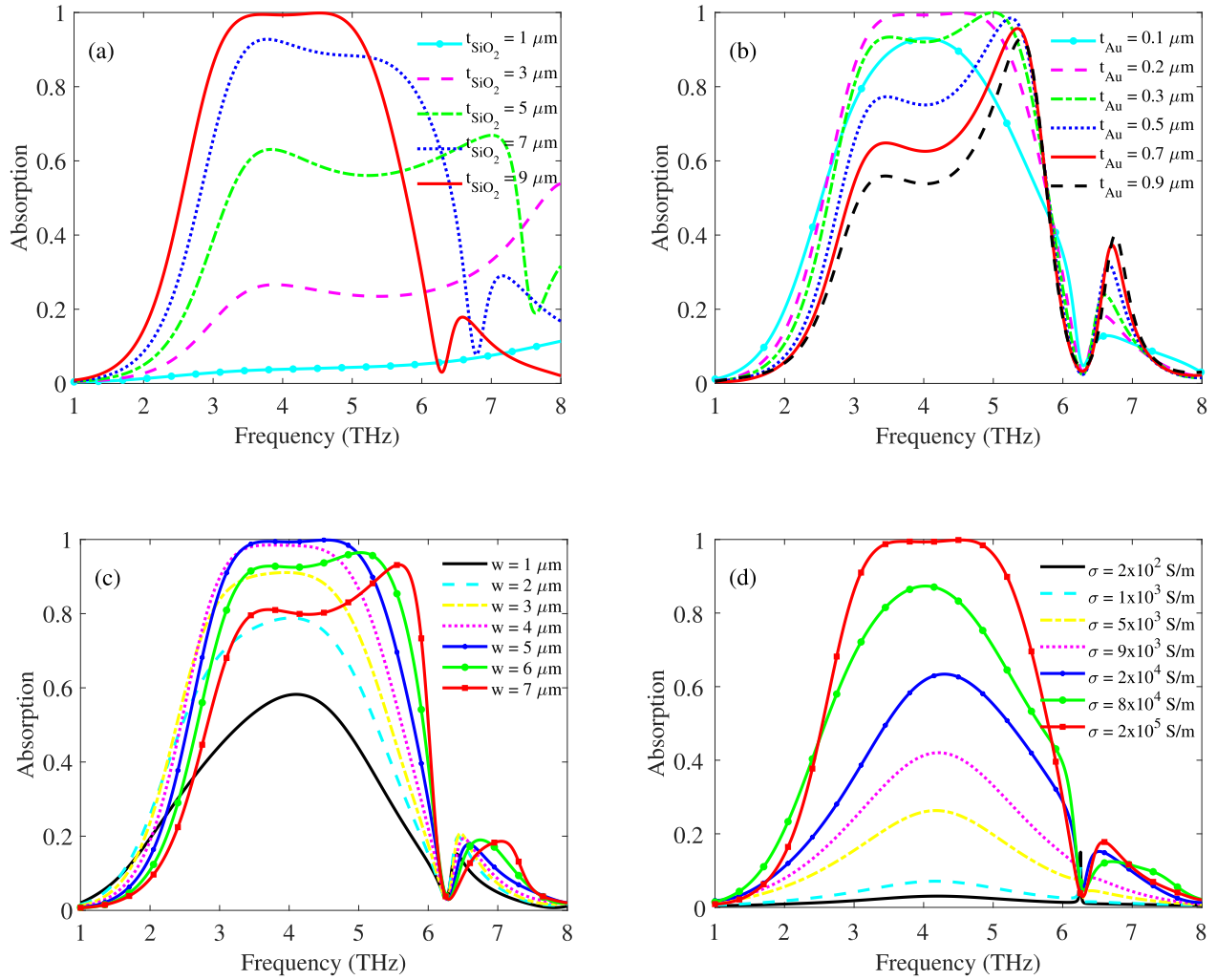


FIGURE 3. Calculated frequency dependent absorptance spectrum on (a) dielectric layer thickness, t_{SiO_2} (b) metal layer thickness, t_{Au} (c) ring width, w and (d) conductivity, σ .

initially improves from 90% at $t_{\text{Au}} = 0.1 \mu\text{m}$ to peak performance at the optimal $0.2 \mu\text{m}$ thickness, beyond which further thickness increments degrade absorption efficiency. The clear identification of this optimal thickness at $0.2 \mu\text{m}$ provides crucial design guidelines for maximizing absorption in this material system, emphasizing that simply increasing material thickness does not necessarily enhance performance and may in fact degrade absorption characteristics beyond the optimal point. Figure 3(c) demonstrates the critical relationship between ring width and absorption performance, revealing a distinct optimization window. As the width increases from $1 \mu\text{m}$ to $7 \mu\text{m}$, the absorption characteristics exhibit a three-stage behavior: initial low performance ($<60\%$ absorption at $1 \mu\text{m}$), followed by peak efficiency ($\approx 100\%$ at the optimal $5 \mu\text{m}$ width), and subsequent degradation (declining to 80% at $6-7 \mu\text{m}$). This non-monotonic response establishes $5 \mu\text{m}$ as the ideal design point, simultaneously maximizing both absorption magnitude (approaching unity) and operational bandwidth. The

maximum performance is observed at a wavelength of $5 \mu\text{m}$, accompanied by reduced efficiency at both smaller and larger widths, provides clear justification for selecting this geometric parameter in our absorber design, as it represents the optimal balance between bandwidth enhancement and absorption intensity.

The critical relationship between VO_2 's conductivity (200 S/m to $2 \times 10^5 \text{ S/m}$) and absorption performance, demonstrating dynamic tunability while maintaining spectral stability is demonstrated in Fig. 3(d). As conductivity increases, the absorptance undergoes a dramatic enhancement from 0.05 to near-unity absorption, with the resonance frequency consistently anchored at 4.5 THz throughout the tuning range. This conductivity-dependent behavior stems from the corresponding modulation of VO_2 's permittivity, which directly governs the electromagnetic interaction. The findings establish conductivity control as a powerful mechanism for precisely tailoring absorption characteristics, enabling real-time performance optimization for diverse

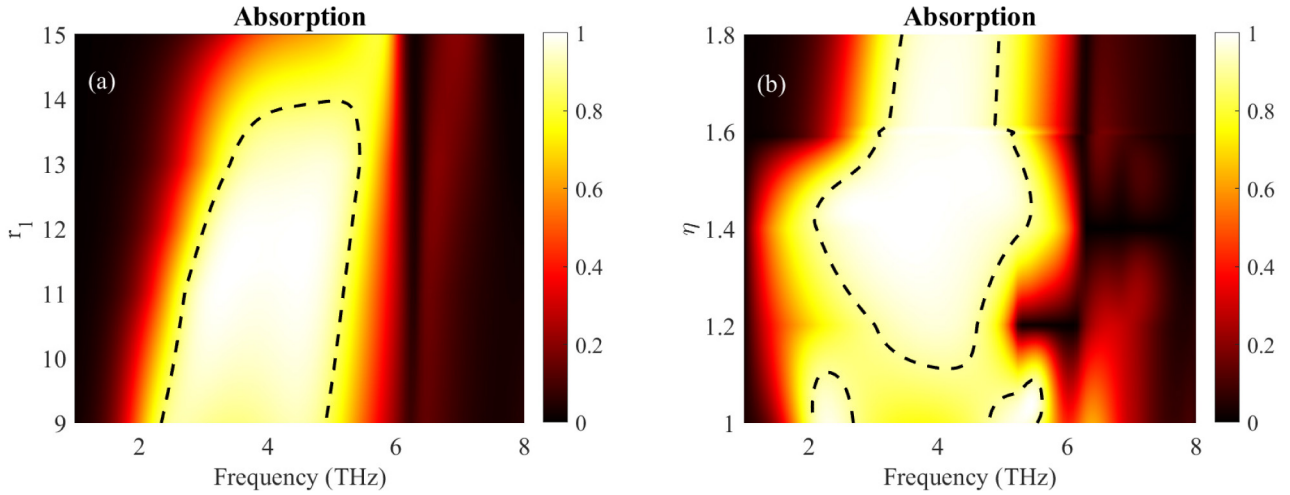


FIGURE 4. Simulated absorption spectrum as a function of (a) r_1 (radius along semi-major axis) and (b) η . In both plots, absorption exceeding 90% is highlighted by black dashed lines.

terahertz applications without altering the device's physical geometry.

Figure 4(a) displays a comprehensive color-mapped analysis of absorption characteristics as a function of both frequency and resonator semi-major axis radius, ranging from 9 μm to 15 μm , where the color gradient quantitatively represents absorption levels from minimal (dark red) to maximal (yellow). The data reveal a distinct optimization pattern: absorption efficiency initially improves with increasing radius, reaches near-perfect absorption ($\approx 100\%$) at the optimal 12 μm configuration, and subsequently declines for larger dimensions. Notably, with 12 μm semi-major axis, the absorber exhibits superior performance across the 2.5-5 THz band, as evidenced by sustained high absorption values exceeding the 90% threshold (marked by the dashed black line). These results demonstrate that the semi-major axis radius critically controls both the magnitude and spectral position of absorption, with 12 μm representing the geometrically optimized condition that balances resonant enhancement with broadband performance in our design.

Figure 4(b) systematically examines how the aspect ratio (ellipticity ratio), η , varying from 1.0 to 1.8 at fixed 5 μm ring width, controls absorption characteristics through a detailed color plot with quantified absorption levels. The data reveal two distinct operational regimes: at $\eta = 1$ (circular resonator), the absorber exhibits dual-band performance with two narrow 90% absorption peaks separated by 2.5 THz, while increasing η transforms the response into a broadband absorber. Optimal performance emerges at $\eta = 1.6$, achieving either perfect 100% absorption across 1.5 THz bandwidth or 98% absorption over an extended 2.3 THz bandwidth, representing the most favorable trade-off between absorption intensity and spectral coverage. This demonstrates precise geometric control via η enables switching between multi-band and broadband absorption modes while maintaining high efficiency.

Figure 5(a) characterizes the angular dependence of absorption performance for incident angles ranging from 0° to 100° (in 20° increments), revealing significant polarization-sensitive behavior attributable to the deliberate asymmetry of the structure, and is calculated for transverse electric (TE)-polarization only. While TE-polarized waves maintain stable absorption up to 40° incidence, performance degrades markedly beyond 70°, with the primary absorption peak diminishing progressively at higher angles. This angular sensitivity stems directly from the elliptical resonator geometry, which introduces orientation-dependent electromagnetic interactions. The absorber demonstrates optimal efficiency at normal and low incident angles ($< 40^\circ$), but its asymmetric design necessarily compromises both wide-angle performance and polarization insensitivity - tradeoffs that could be mitigated in symmetrical configurations but were intentionally accepted in this design to achieve other targeted performance characteristics.

Figure 5(b) systematically evaluates the polarization-angle dependence of absorption performance ranging from 0° to 100°, demonstrating how structural asymmetry in the elliptical design inherently limits polarization insensitivity. The results reveal two stable operational regimes: consistent absorption spectra from 0° to 40° and from 60° to 100°, where similar resonant peaks are maintained, contrasted by a distinct transition zone between 40°–60°, showing measurable absorption variations. This behavior directly correlates with the elliptical resonator's geometric anisotropy, which creates polarization-dependent electromagnetic interactions. While the structure maintains functional broadband absorption across most polarization angles, the observed 40°–60° transition region quantitatively confirms that perfect polarization independence cannot be achieved with this asymmetric design – a fundamental tradeoff inherent to elliptical metamaterial absorbers that must be balanced against their spectral and tunability advantages.

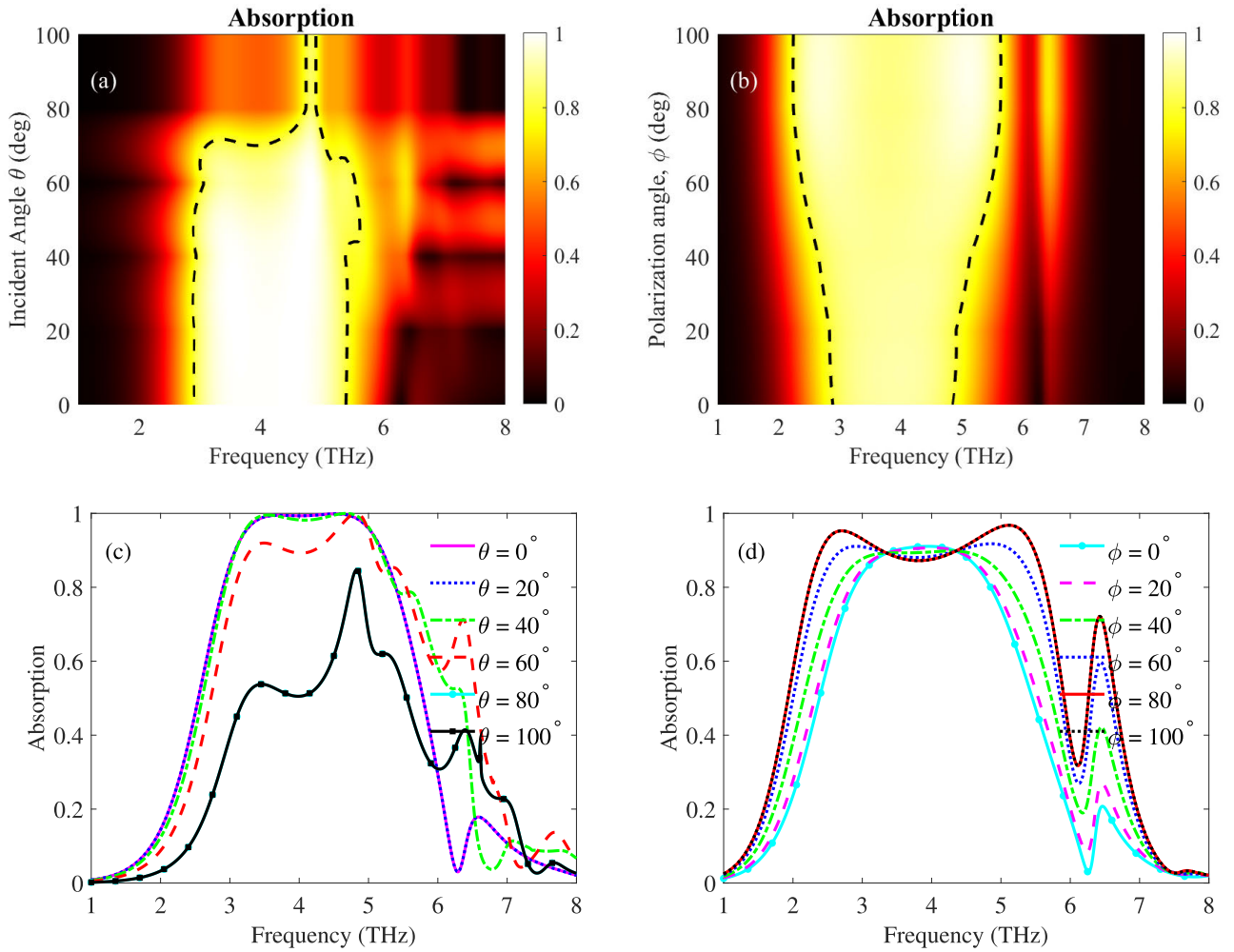


FIGURE 5. Simulated absorption spectrum as a function of (a) incident angle, θ and (b) polarization angle, ϕ . In both plots, absorption exceeding 90% is highlighted by black dashed lines. Angular sensitivity of the absorber's performance for (c) incident angle, θ and (d) polarization angle, ϕ .

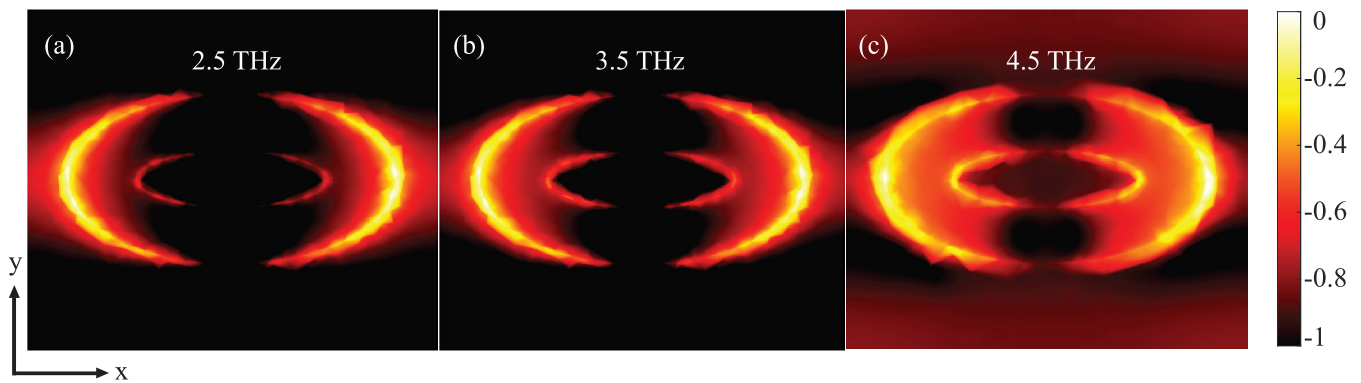


FIGURE 6. Normalized electric field distribution (log scale) at (a) 2.5 THz, (b) 3.5 THz and (c) 4.5 THz, respectively.

Figures 5(c) and (d) present the absorption spectra as a function of frequency under varying incident and polarization angles, simulated exclusively for TE polarization. To comprehensively assess angular sensitivity, the incident angle was systematically incremented in steps of 20° from

normal incidence (0°) up to a highly oblique angle of 100°. As shown in Fig. 5(c), absorptivity exhibits the expected inverse dependence on the incident angle, decreasing as the angle increases. However, analysis of the dependence of the polarization angle reveals a trade-off: the absorption

bandwidth increases with the polarization angle, at the expense of a reduction in peak absorptivity, which is shown in Fig. 5(d).

Figure 6 analyzes the electric field distribution across three characteristic frequencies (2.5 THz, 3.5 THz, and 4.5 THz), revealing distinct localization patterns that correlate with absorption performance. At 2.5 THz, as shown in Fig. 6(a), the field concentrates weakly on the ring's inner surface, while the peak absorption frequencies, *i.e.*, at 3.5 THz and 4.5 THz show progressively stronger field enhancement - particularly at 4.5 THz, as indicated in Fig. 6(c), where the field uniformly distributes across the entire ring structure, explaining the maximum absorption efficiency. This frequency-dependent field confinement directly demonstrates how geometric resonance effects govern the absorption mechanism.

Furthermore, a comparative analysis with recent VO₂-based absorbers (Table 2) confirms our design's superior performance in bandwidth and tunability, establishing its advancements in metamaterial absorber technology.

TABLE 2. Comparison among recently published papers based on VO₂.

References	Absorptance (%)	Bandwidth (THz)	Tunable Range
[21]	> 90	1.2	—%
[43]	> 90	0.3	20%-98%
[44]	> 90	1.93	—
[45]	> 90	2.18	32.8%-99.9%
[46]	> 90	1.40	—
[47]	> 90	2.30	2%-100%
Proposed Model	≅ 100	1.5	0.05%-100%
	> 90	2.11	0.05%-100%

A. TOLERANCE ANALYSIS AND MODEL VERIFICATION

To account for potential variations in geometric parameters during fabrication, a tolerance analysis was performed. We investigate tolerance analysis by systematically varying critical design parameters—including the unit cell's aspect ratio (η), dielectric spacer thickness (t_{SiO_2}), metal thickness (t_{Au}) and lattice period (p)—around their optimal values. A comprehensive sensitivity analysis was performed by applying a systematic $\pm 3\%$ perturbation to each parameter individually, a method designed to emulate the range of typical manufacturing deviations and tolerances. The key results of this analysis, which quantify the impact of each variation on the performance of the system, are summarized in Fig. 7. As shown in Figs. 7(a-d), the absorber demonstrates remarkable robustness to manufacturing errors within a $\pm 3\%$ range of the optimal values. The tolerance analysis revealed that the system's performance is largely insensitive to variations in its geometrical parameters, as they demonstrate a minimal response to deviations from their optimal values. Crucially, the absorption bandwidth remains highly stable, varying by less than 1% under all conditions. These

findings confirm the design's high tolerance to fabrication imperfections and its ability to maintain reliable performance despite manufacturing variations.

We now validate the results of the CST simulation by comparing them with the finite element-based software COMSOL simulations. Note that the boundary conditions in COMSOL are configured analogously to the CST model. The structure is illuminated by an electromagnetic wave propagating in the z -direction, which is excited by a Floquet port assigned to its upper surface. To maximize simulation accuracy, the model was discretized using an extremely fine mesh. As shown in Fig. 8, the absorption spectra of both solvers exhibit excellent agreement in both operational bandwidth and near-unity absorption efficiency. This consistency conclusively validates the reliability of the performance of the metamaterial absorber.

IV. ML-BASED PARAMETER PREDICTION

ML-Based Parameter Prediction involves the use of machine learning algorithms to determine optimal parameter values to achieve specific outcomes, such as target absorbance levels in material design. This approach is especially useful in photonics and material science, where the relationship between structural parameters and optical properties is often highly complex and nonlinear. Traditional design approaches typically rely on time-consuming, iterative experimentation, whereas ML can efficiently analyze large datasets, uncover intricate patterns, and predict the best parameters for desired performance. Furthermore, ML enables inverse design—starting from a target outcome (e.g., a particular absorbance spectrum) to derive the required parameters—dramatically speeding up the design process and enhancing efficiency. Recent studies highlight the integration of ML with MMAs for applications in THz and microwave frequencies. One study presents a multi-band MMA for gas sensing that utilizes concentric octagonal ring resonators; ML is employed to optimize the structure's performance [48]. Another study evaluated multiple ML models, identifying the Extra Trees regressor as the most accurate for predicting absorption in THz range [49]. The comparison of ML models such as XGBoost and LightGBM for predicting MMA performance in the microwave spectrum [50], alongside other works, demonstrates the considerable potential of machine learning to optimize MMA designs across a range of applications. The predictive analysis in this work utilizes multiple models. The following section details the essential data preparation stages performed prior to model training.

A. DATASET ACQUISITION AND PRE-PROCESSING

The dataset used in this study was extracted from a structured CSV file containing input features and their corresponding output parameters for a physical design task. After loading the data into a pandas DataFrame, it was inspected for completeness and consistency. The dataset comprises 700 entries with 11 columns, where r_1 , t_{SiO_2} , t_{Au} , w ,

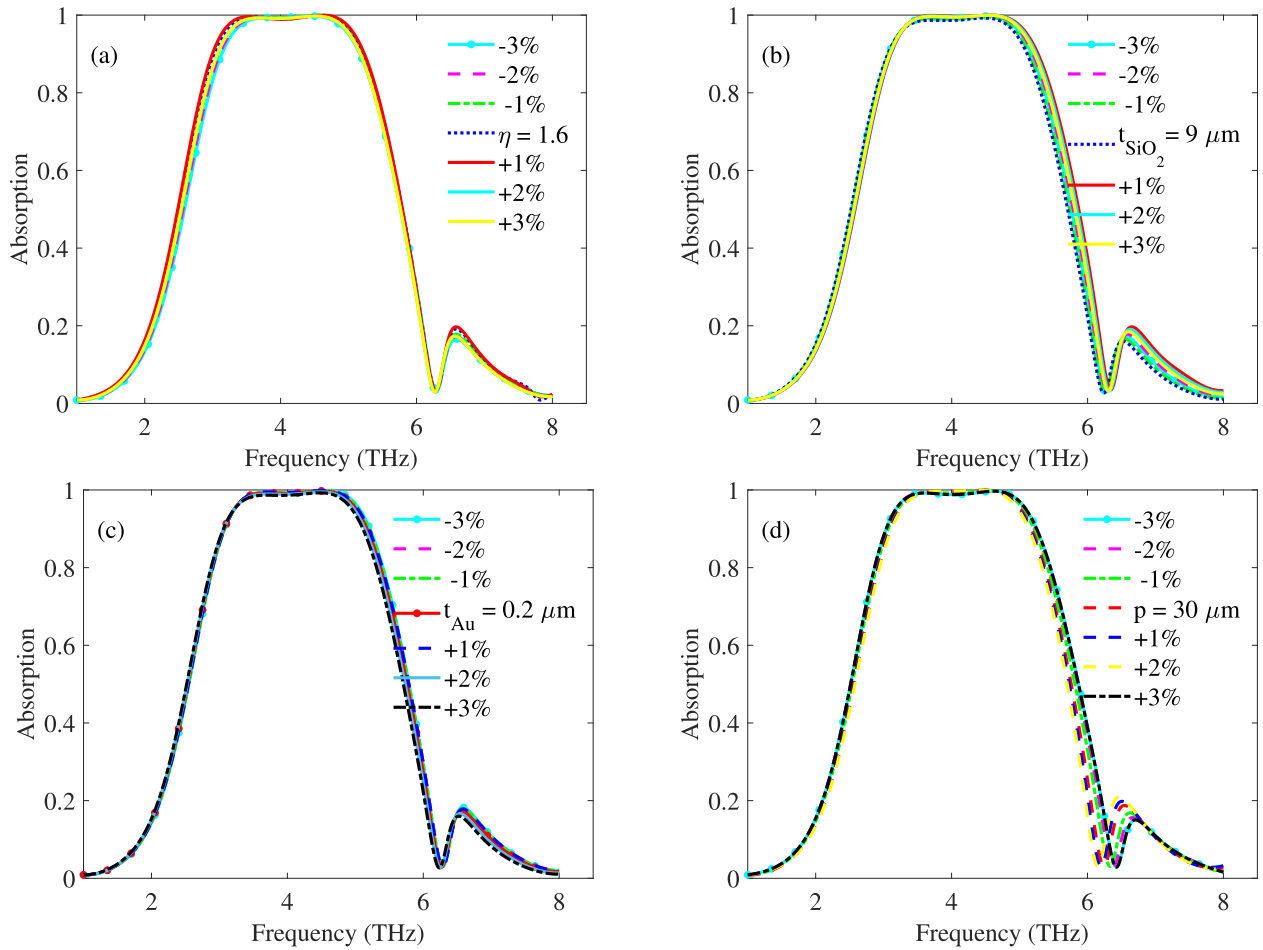


FIGURE 7. The impact of simulated manufacturing errors on the absorption spectra with the variation of (a) η , (b) t_{SiO_2} , (c) t_{Au} and (d) p .

TABLE 3. Dataset collected from the CST simulation.

r_1	w	η	t_{SiO_2}	t_{Au}	peak_freq	peak_abs	bandwidth	vswr	resonance_loss	imp_matching
11.80	4.90	1.59	9.30	0.20	4.89	0.99	1.80	1.98	17.46	4.52
12.00	5.00	1.60	9.00	0.20	4.50	0.99	1.84	1.08	28.46	1.08
12.10	5.10	1.58	8.90	0.20	4.79	0.91	1.97	1.35	47.93	4.70
12.00	3.70	1.60	9.00	0.20	3.95	0.97	1.47	1.44	14.94	1.44
12.00	5.00	1.60	7.40	0.20	3.95	0.96	2.21	1.53	13.63	1.53
11.90	1.10	1.58	12.00	0.20	5.35	0.93	0.34	1.35	12.85	1.66
-	-	-	-	-	-	-	-	-	-	-

and η serve as the predicted parameters, while the remaining columns function as input features.

B. DATA AUGMENTATION USING CTGAN

To address the challenge of limited data and improve the generalization of the model, synthetic samples were generated using a Conditional Tabular Generative Adversarial Network (CTGAN) from the SDV library. The CTGAN model was trained on the original dataset, producing 500 synthetic samples that preserved the statistical properties of the real data while expanding the range of design parameters. This

augmentation increased the total dataset size to 1200 samples, improving both diversity and robustness for training.

1) VALIDATION OF CTGAN DATA AUGMENTATION

The efficacy of the CTGAN-based data augmentation was rigorously validated through distribution similarity metrics and an ablation study. The synthetic data demonstrated high fidelity to the original dataset, with an average Jensen-Shannon Divergence of 0.048 and Wasserstein Distance of 0.082 across all features (Table 4), confirming that the generative model successfully learned the underlying data manifold.

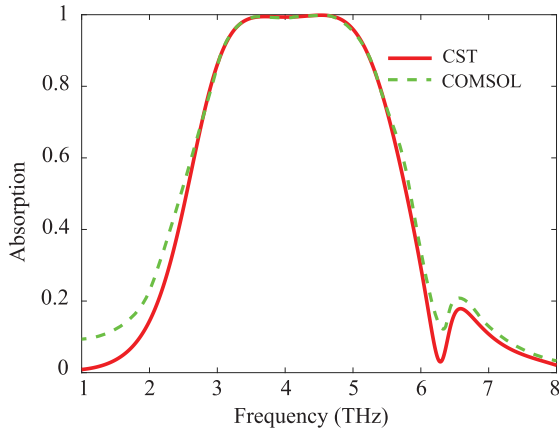


FIGURE 8. Calculated absorption spectra: A cross-verification between CST (red) and COMSOL (dotted green).

C. FEATURE ENGINEERING AND CORRELATION ANALYSIS

Following data augmentation, we engineered six new features—`log_bandwidth`, `bandwidth_squared`, `vswr_squared`, `imp_matching_squared`, `bandwidth_vswr`, and `imp_matching_bandwidth`—derived from the original inputs (Table 5). Correlation matrices were analyzed to assess variable relationships, and redundant or low-relevance features were eliminated to minimize noise and improve model interpretability. The refined dataset was then normalized and preprocessed for training.

D. FEATURE SELECTION

The correlation analysis guided the selection of features, retaining only those with significant predictive value to avoid redundancy and irrelevant data. This step was crucial for optimizing model performance. In this study, distinct feature sets were tailored for each target variable, ensuring the selected inputs accurately capture the underlying relationships that influence predictions, thereby enhancing model accuracy.

E. PREDICTIVE MODELING

This regression task aimed to predict multiple output variables from a set of input features. The primary model, a Random Forest Regressor, was implemented within a MultiOutputRegressor framework to handle simultaneous multi-target predictions. To ensure robustness, the dataset was split into 80% training and 20% testing sets. Performance was further evaluated using a suite of machine learning models, ranging from classical techniques to advanced methods, for a comprehensive comparison which are Random Forest Regressor (RF), HistGradientBoosting Regressor (Hist), CatBoost Regressor (CAT), XGBoost Regressor (XG), LightGBM Regressor (Light), Linear Regression (LR), Decision Tree Regressor (DT), ElasticNet Regressor (EN), Ridge Regression (Ridge), Lasso Regression (Lasso), Neural Networks (NN). To address the multi-objective optimization challenge inherent in the system, a hybrid computational framework integrating

TABLE 4. Distribution similarity metrics (Original vs. Synthetic Data).

Feature	Distribution Similarity Metrics	
	Wasserstein Distance	Jensen-Shannon Divergence
r_1	0.152	0.062
w	0.088	0.045
η	0.095	0.078
$\vdots$	$\vdots$	$\vdots$
Mean (All Features)	0.082	0.048

a genetic algorithm with a machine learning surrogate model was developed. The process commenced with a Non-dominated Sorting Genetic Algorithm II (NSGA-II) to identify a Pareto-optimal front of solutions. The five target variables (r_1 , w , η , t_{SiO_2} , t_{Au}) were defined as minimization objectives. A population of 100 individuals, each representing a candidate solution vector of input features, were evolved over 50 generations. Genetic operators, including simulated binary crossover (cxBlend) and Gaussian mutation, were employed to ensure a diverse and convergent search.

F. INVERSE DESIGN AND 2D VISUALIZATION

After training the model, we visualized predictions in a 2D space to analyze how input feature variations influenced output parameters. This visualization effectively demonstrated the inverse design process, enabling us to determine optimal input values based on desired results. We then implemented these predicted parameters in CST simulations to verify their accuracy by comparing modeled absorbance and other properties with actual outcomes, ensuring alignment between predictions and expected performance.

V. RESULTS AND ANALYSIS

This study evaluated multiple machine learning models for predicting target parameters (r_1 , w , η , t_{Au} , and t_{SiO_2}) using Mean Squared Error (MSE) and R-squared (R^2) metrics. The Random Forest Regressor achieved the highest R^2 values for r_1 , w , η , and t_{SiO_2} , demonstrating superior predictive performance. CatBoost showed consistent results, particularly for t_{Au} ($R^2 = 0.57$), while XGBoost and LightBoost delivered comparable but generally inferior performance to both Random Forest and CatBoost across all targets. MSE values closer to 0 indicate better performance, whereas R^2 values closer to 1 suggest that the model explains a larger proportion of the variance in the data. The results are summarized in Table 6.

Conventional parameter optimization typically relies on an inefficient trial-and-error approach, requiring designers to manually adjust each parameter and evaluate performance through repeated simulations or physical experiments. This iterative process proves both time-consuming and computationally expensive, often demanding numerous cycles before achieving optimal results. ML provides a data-driven

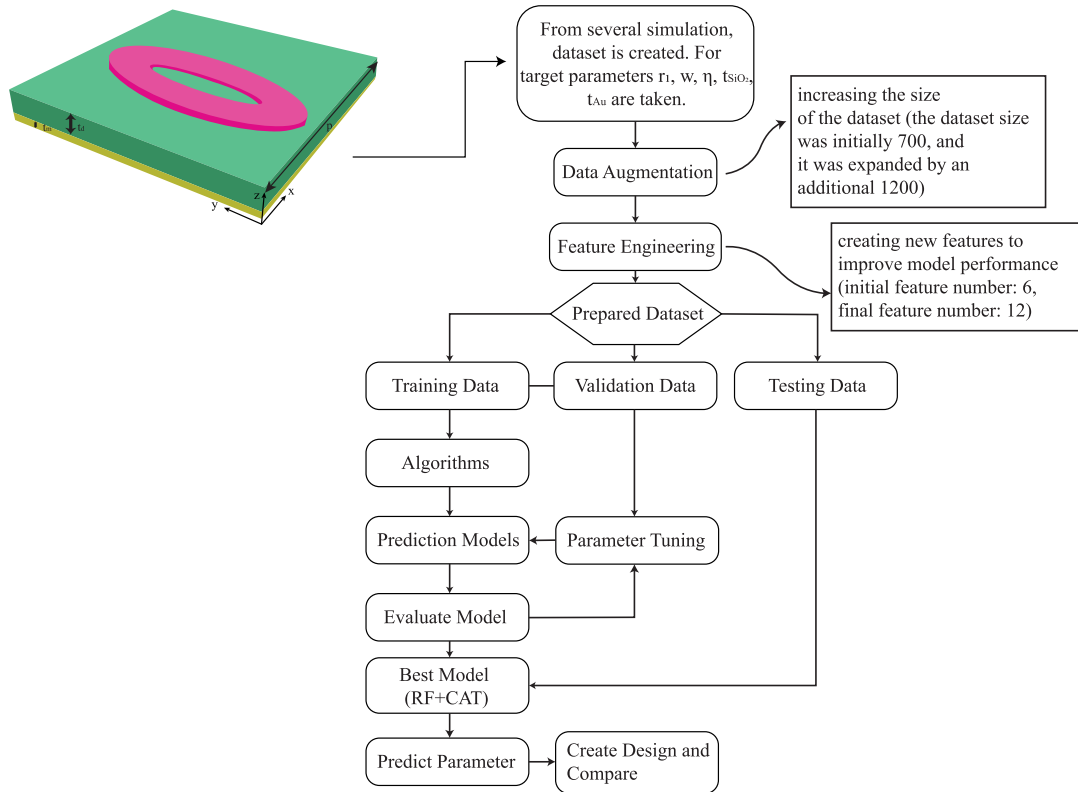


FIGURE 9. The figure outlines the sequential steps starting from data augmentation, feature engineering, and dataset preparation, followed by the creation of training, validation, and testing datasets. Various algorithms and prediction models are applied, followed by parameter tuning and model evaluation to determine the best model (RF+CAT). Finally, the selected model is used to predict parameters, which are then compared and used for design creation.

TABLE 5. New features after feature engineering.

log_bandwidth	bandwidth_squared	vswr_squared	imp_matching_squared	bandwidth_vswr	imp_matching_bandwidth
1.03	3.25	3.93	20.42	3.58	8.15
1.04	3.39	1.16	1.16	1.99	1.99
1.09	3.86	1.83	22.06	2.66	9.23
0.91	2.17	2.06	2.06	2.12	2.12
1.17	4.89	2.33	2.33	3.37	3.37

TABLE 6. Feature columns for each target column.

Target Column	Feature Columns
r_1	peak_freq, peak_abs, bandwidth, vswr
w	peak_freq, vswr_squared, imp_matching_squared, log_bandwidth, bandwidth_vswr
η	peak_freq, peak_abs, bandwidth, vswr, resonance_loss, imp_matching, vswr_squared, imp_matching_squared, log_bandwidth, bandwidth_squared, bandwidth_vswr
t_{Au}	peak_freq, bandwidth, resonance_loss, imp_matching
t_{SiO_2}	peak_freq, peak_abs, bandwidth, resonance_loss, imp_matching_bandwidth

alternative to traditional absorber design by establishing predictive relationships between structural parameters and performance metrics. Through training on historical absorber configurations and their corresponding characteristics (e.g., bandwidth, absorptance), ML models learn to accurately predict performance outcomes. These trained models enable

rapid parameter optimization, significantly reducing the iterative cycles required in conventional approaches. This transformation from manual trial-and-error to automated prediction not only enhances design efficiency but also improves performance predictability while conserving computational resources.

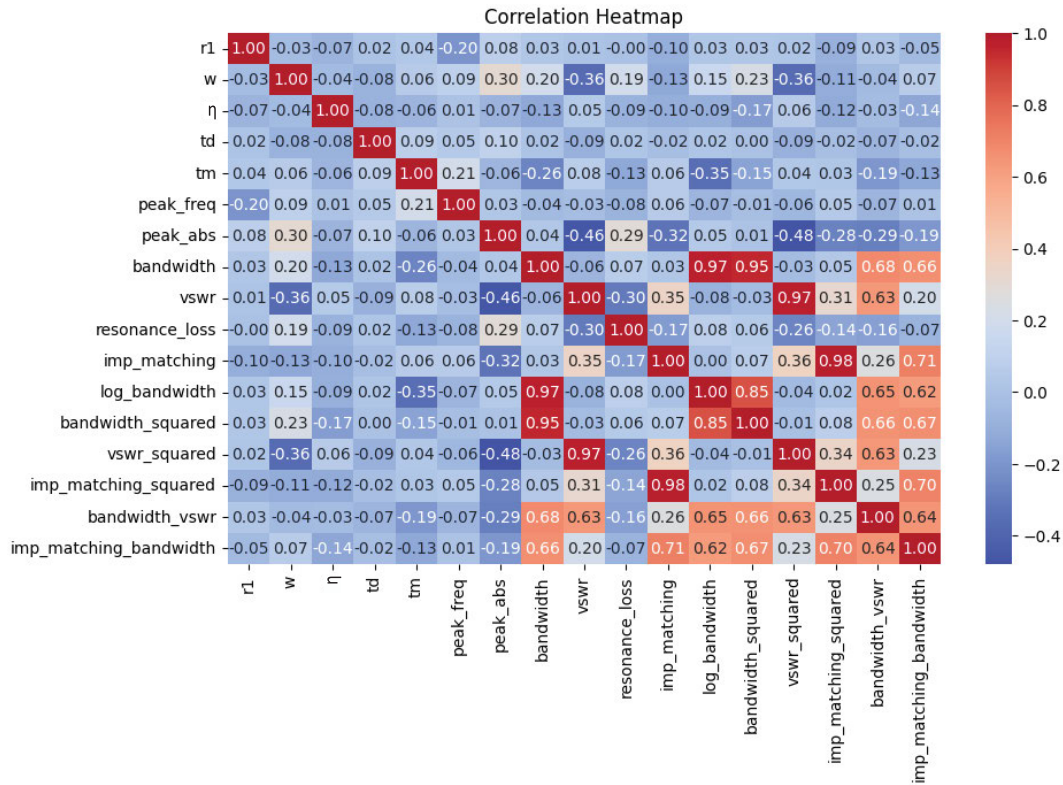


FIGURE 10. Correlation heatmap illustrating the relationships between various parameters. The matrix shows the correlation coefficients between different features, with values ranging from -1 (strong negative correlation) to 1 (strong positive correlation). The color scale on the right indicates the strength and direction of the correlations, with red representing positive correlations and blue indicating negative correlations.

TABLE 7. Model evaluation results for each target column.

Model	r_1		w		η		t_{Au}		t_{SiO_2}	
	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2
RF	0.31	0.63	0.29	0.65	0.01	0.51	0.04	0.56	0.02	0.64
Hist	0.37	0.33	0.31	0.41	0.02	0.008	0.0572	0.491	0.02	0.008
CAT	0.3247	0.4206	0.293	0.4597	0.0182	0.1371	0.0474	0.5775	0.0182	0.1371
XG	0.334	0.403	0.3709	0.3177	0.0186	0.1174	0.0941	0.1621	0.0186	0.1174
Light	0.38	0.309	0.34	0.37	0.02	0.03	0.05	0.523	0.02	0.03
LR	0.56	-0.0015	0.52	0.0277	0.0213	-0.012	0.0998	0.115	0.0213	-0.012
DT	0.65	-0.16	0.85	-0.58	0.04	-1.21	0.044	0.601	0.04	-1.21
EN	0.56	-0.01	0.53	0.022	0.0211	-0.002	0.11	-0.0096	0.0211	-0.002
Ridge	0.55	0.011	0.53	0.02	0.021	-0.014	0.099	0.111	0.021	-0.014
Lasso	0.56	-0.01	0.55	-0.008	0.02	-0.002	0.11	-0.009	0.02	-0.002
NN	0.69	-0.23	0.43	0.2	0.02	-0.27	0.05	0.48	0.02	-0.27

The dataset used for training the model is derived from the simulation of the absorber system shown in Fig. 1. The dataset consists of 11 columns, where the columns r_1 , w , η , t_{SiO_2} , and t_{Au} are the target variables. These correspond to the semi-major axis of the ring r_1 , thickness of SiO_2 (t_{SiO_2}), the thickness of Au (t_{Au}), the ratio of semi-major to semi-minor axis (η) and the width of the ring (w). These parameters are predicted by the model. The remaining six columns—peak_freq, peak_abs, bandwidth, vswr, resonance_loss, and imp_matching act as the feature columns for the model,

representing key performance metrics related to the absorber. MSE and R^2 are widely regarded as the most effective and commonly used evaluation metrics for regression problems. The Table 7 presents the performance of various machine learning models (RF, Hist, CAT, XG, Light, LR, DT, EN, Ridge, Lasso, NN) on multiple target columns (r_1 , w , η , t_{SiO_2} , and t_{Au}) using MSE and R^2 metrics. While some models, such as RF and CAT, perform relatively well across several targets, others, like LR and DT struggle, showing negative R^2 values for some targets, which indicates poor model fit. RF shows

TABLE 8. Comparison of predicted and actual values for input parameters.

Input Parameters	Predicted Values	Actual Values	% of Error	CI Coverage
Peak Frequency: 4.49 THz	r_1 : 11.5 μm	11.8 μm	2.5	0.93
Bandwidth: 1.8 THz	w : 4.5 μm	4.9 μm	8.1	0.92
Peak Abs: 0.99	η : 1.61	1.59	1.2	0.93
VSWR: 1.98	t_{Au} : 0.16 μm	0.17 μm	20	0.92
Resonance Loss: 17.46	t_{SiO_2} : 9.1 μm	9.3 μm	2.1	0.93
Impedance Matching: 4.52	—	—	—	—
Peak Frequency: 2.61 THz	r_1 : 12.03 μm	12.1 μm	0.57	0.93
Bandwidth: 1.9 THz	w : 4.8 μm	5 μm	4	0.92
Peak Abs: 0.95	η : 1.57	1.59	1.25	0.93
VSWR: 1.46	t_{Au} : 0.21 μm	0.2 μm	4.76	0.92
Resonance Loss: 47.96	t_{SiO_2} : 9.1 μm	9.2 μm	1.08	0.93
Impedance Matching: 1.82	—	—	—	—

TABLE 9. Tolerance analysis of ML-predicted geometric parameters.

By Changing Parameters	Output with Changing Predicted Values
r_1 : 12 w : 5 η : 1.6 t_{Au} : 9 t_{SiO_2} : 0.2	Peak Absorbance: 0.99 Bandwidth: 1.84
r_1 : 11.9 w : 4.9 η : 1.5 t_{Au} : 8.9 t_{SiO_2} : 0.1	Peak Absorbance: 0.99 Bandwidth: 2.14
r_1 : 12.1 w : 5.1 η : 1.7 t_{Au} : 9.1 t_{SiO_2} : 0.3	Peak Absorbance: 0.96 Bandwidth: 2.28

the best performance for r_1 , w , η , t_{SiO_2} . CAT performs well for t_{Au} but poorly for others. Overall, the results suggest that different models may be better suited for different target columns, with room for further optimization, particularly in cases with negative R^2 values.

Table 8 compares the predicted and actual values for various input parameters and calculates the percentage error for each at a specific set of input parameters. For example, the predicted value (11.5 μm) for semi-major axis differs from the actual value of 11.8 μm , with a small error of 2.5% at peak frequency of 4.49 THz. The ring width prediction is significantly off from the actual value of 4.9 μm , resulting in an 8.1% error. The predicted ratio (1.61) is higher than the actual (1.59), but the error is relatively small at 1.2%. The metal thickness prediction is notably lower than the actual value of 0.17 μm , causing a larger error of 20%. While the prediction error for t_{Au} is higher (20% MAE), our uncertainty quantification demonstrates that the model correctly captures this inherent difficulty. This is validated by the well-calibrated coverage probability of 92%, close to the ideal 95%. Therefore, the model does not make overconfident predictions for t_{Au} but rather provides a useful estimate with a true representation of its uncertainty, which is valid for guiding the optimization process where relative comparisons

TABLE 10. Verification of predicted parameters compared to initial input values.

Feature	Output with Predicted Parameters	Initial Input Value
Peak Frequency	4.03	4.49
Peak Absorbance	0.98	0.99
Bandwidth	1.98	1.80
VSWR	1.75	1.98
Resonance Loss	18.96	17.46
Impedance Matching	3.25	4.52

are key. The prediction of dielectric thickness is slightly lower than the actual value of 9.3 μm , with a 2.1% error. Overall, the table highlights the differences between predicted and actual values with varying degrees of error across parameters.

Table 9 presents the tolerance analysis of the ML-predicted geometric parameters to evaluate the stability of the metamaterial's performance against small variations in its structural dimensions. Three sets of design parameters were analyzed by slightly modifying the geometric features. Despite these dimensional deviations (within ± 0.1 – 0.3 μm), the peak absorbance remains almost unchanged — varying only between 0.96 and 0.99. This demonstrates that the ML-predicted structure maintains a high and stable absorption efficiency, even with fabrication-induced variations. However, the bandwidth shows a moderate increase from 1.84 THz to 2.28 THz, indicating that the resonance broadens slightly as the parameters change. These results confirm that the proposed design exhibits strong fabrication tolerance and structural robustness, ensuring consistent optical performance even when practical manufacturing deviations occur. This further validates the reliability of the ML-predicted geometry for real-world implementation in mid-infrared metamaterial absorbers.

Table 10 presents a comparison between the output obtained using predicted parameters and the initial input values for various features. The Peak Frequency shows a predicted output of 4.03 compared to the initial input of 4.49. Similarly, the Peak absorbance has a predicted value of 0.98, very close to the initial input of 0.99. The Bandwidth output with predicted parameters is 1.98, matching the initial input value of 1.8. The VSWR prediction is 1.75, slightly lower than the initial input value of 1.98. The Resonance Loss

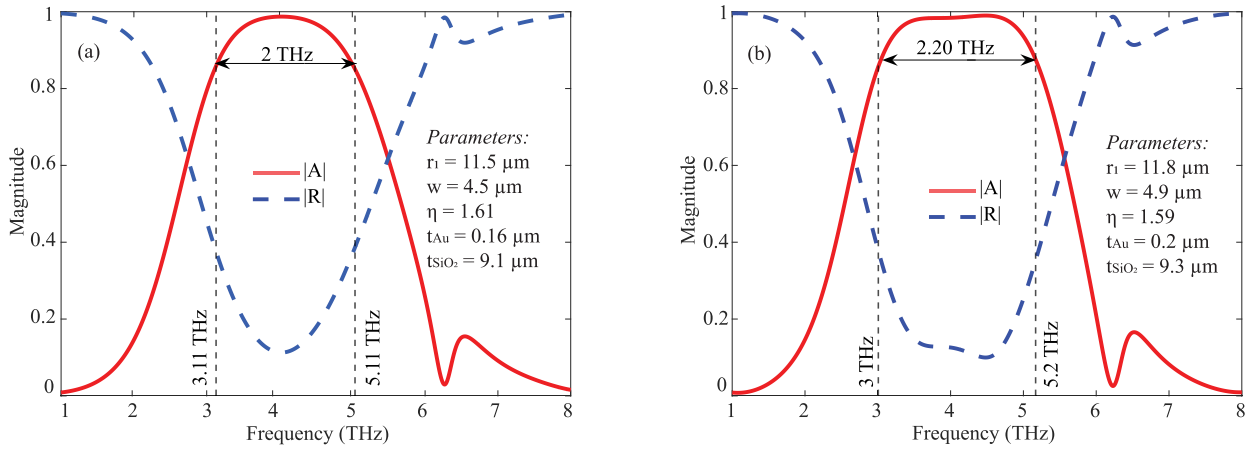


FIGURE 11. Simulated absorption spectrum as a function of frequency for (a) predicted parameters and (b) actual parameters.

predicted value is 18.96, which is higher than the initial input of 17.46. Lastly, Impedance Matching shows a predicted value of 3.25, compared to the initial value of 4.52. This table helps to assess the accuracy of the predictions in relation to the initial input values, with most predictions closely aligning with the initial inputs. Figure 11 compares the absorption spectra for two different sets of parameters, where both plots display the absorption ($|A|$) as the red curve and the reflectance ($|R|$) as the blue dashed curve. In Fig. 11(a), the parameters ($r_1 = 11.5 \mu\text{m}$, $w = 4.5 \mu\text{m}$, $\eta = 1.61$, $t_{Au} = 0.16 \mu\text{m}$ and $t_{SiO_2} = 9.1 \mu\text{m}$) are the model's predicted values, while Fig. 11(b) uses the actual dataset parameters ($r_1 = 11.8 \mu\text{m}$, $w = 4.9 \mu\text{m}$, $\eta = 1.59$, $t_{Au} = 0.2 \mu\text{m}$, and $t_{SiO_2} = 9.3 \mu\text{m}$). The comparison aims to verify the accuracy of the predicted parameters, and both plots show very similar absorption profiles. In Fig. 11(a), the absorption bandwidth is approximately 2 THz, whereas in Fig. 11(b), it is around 2.20 THz with peak absorption occurring at 4.49 THz. Despite this slight frequency shift, the absorption behavior remains nearly identical in both cases, with peak absorption values of 0.98 for the predicted parameters and 0.99 for the actual dataset. This close agreement confirms that the predicted parameters are reasonably accurate as they produce almost indistinguishable absorption characteristics. The ablation study (Table 11) conclusively demonstrates the substantial benefit of data augmentation for model performance across all target variables. The augmented models consistently outperformed their baseline counterparts, with R^2 scores increasing by significant margins—most notably for η , where performance more than doubled from 0.23 to 0.51. This improvement was accompanied by dramatic reductions in mean squared error, particularly for η and t_{Au} . These results confirm that the augmentation strategy effectively enhances the model accuracy and generalization, providing more reliable predictions for the optimization task. For further verification, we have performed K-cross validation on the data set, which is shown in Table 12. The model exhibited stable performance in all partitions,

TABLE 11. Model performance comparison between baseline and augmented training data.

Target	Training Data	R^2	MSE
r_1	Baseline (Original)	0.55	0.405
	Augmented	0.63	0.310
w	Baseline (Original)	0.55	0.118
	Augmented	0.65	0.290
η	Baseline (Original)	0.23	0.170
	Augmented	0.51	0.010
t_{Au}	Baseline (Original)	0.38	0.205
	Augmented	0.56	0.040
t_{SiO_2}	Baseline (Original)	0.58	0.045
	Augmented	0.64	0.020

TABLE 12. Results of 5-fold cross-validation.

	Target	MSE	R^2
K Fold Cross Validation k = 5	r_1	0.3159	0.4364
	w	0.2900	0.4500
	η	0.0100	0.2100
	t_{Au}	0.0400	0.5600
	t_{SiO_2}	0.3500	0.4100

confirming its generalizability. The narrow confidence intervals for the performance metrics further exclude the possibility of overfitting. Figure 12 provides a quantitative diagnostic of the model's predictive accuracy by comparing the residual distributions across five output parameters. Each subplot of Fig. 12 (a-e) displays the error distribution for a specific parameter, visualized as a histogram with a kernel density estimate (KDE) curve. Each distribution would be normally distributed with a mean value approximating zero, indicating unbiased predictions, and a small standard deviation, signifying high precision. The shape and range of these distributions allow for a direct numerical comparison of model performance, revealing which parameters were

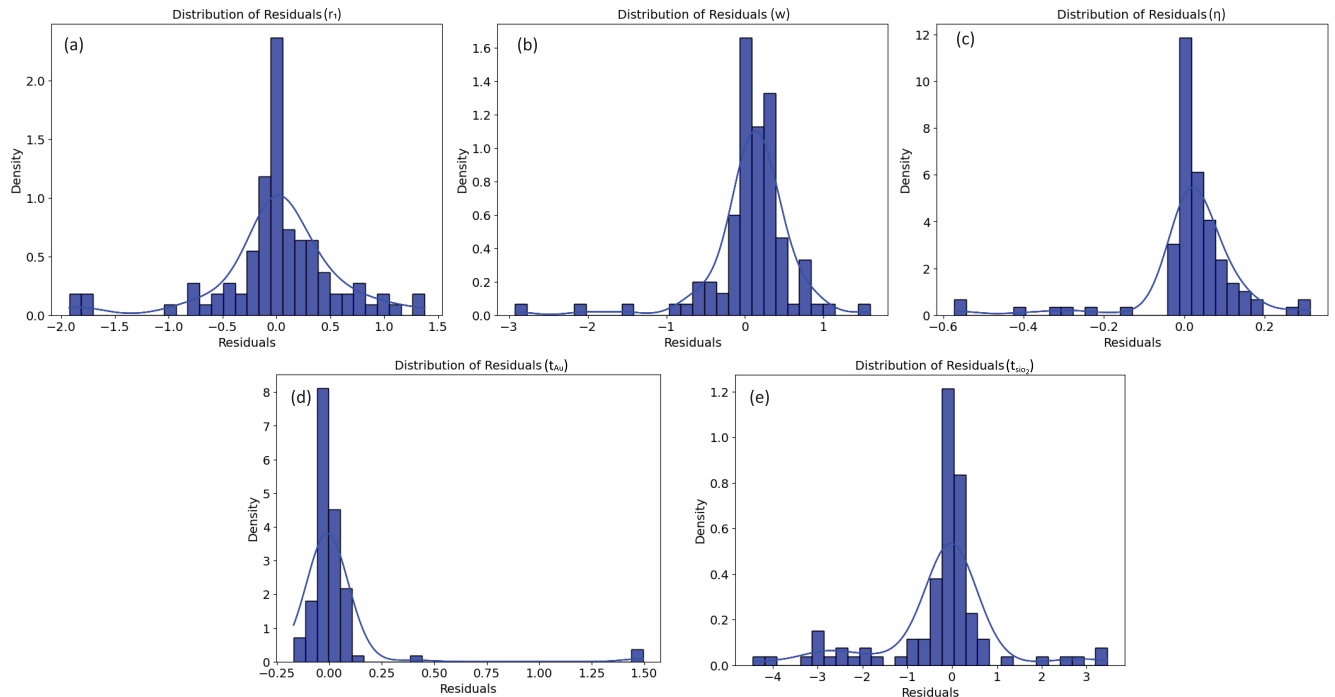


FIGURE 12. Distribution of model prediction residuals (error) for the five target parameters: (a) r_1 , (b) w , (c) η , (d) t_{Au} , and (e) t_{SiO_2} .

more challenging to predict accurately and thus have a wider dispersion of residuals.

VI. POSSIBLE FABRICATION METHOD

One of the key advantages of the proposed absorber is the simplicity of its fabrication process, making it suitable for practical applications. The practical fabrication of the proposed single-layer VO₂ elliptical ring structure, while simpler than multi-layered alternatives, presents specific challenges centered on the VO₂ layer: depositing a high-quality film and patterning it into precise resonators.

First, the deposition of VO₂ films with a sharp insulator-to-metal transition (IMT) is well-established through techniques such as pulsed laser deposition (PLD) and magnetron sputtering method [51], [52], [53]. However, achieving this on the chosen dielectric substrate (e.g., SiO₂) may require careful optimization of temperature and Oxygen's partial pressure to ensure the correct stoichiometry and crystallinity. The most significant challenge lies in the high-resolution patterning of the VO₂ film into the elliptical ring [54]. Standard photolithography followed by plasma etching could be used, but it requires a robust etch mask and precise control to achieve smooth sidewalls and maintain the critical dimensions of the ring width, which is essential for the predicted resonant response. Alternatively, focused ion beam (FIB) milling offers high precision and is excellent for prototyping but has low throughput and can cause surface damage [55]. For larger-scale fabrication, nano-imprint lithography could be a promising future direction [56].

Finally, the deposition of the bottom metal ground plane is a relatively straightforward step. The potential misalignment between the VO₂ pattern and the ground plane is expected to have a minimal impact on performance, as the absorption mechanism is dominated by the resonator's interaction with the incident wave and the ground plane primarily serves as a back-reflector.

VII. CONCLUSION

In this paper, we propose an ultra-broadband metamaterial absorber that achieves over 90% absorptance within a 2.11 THz bandwidth under normal incidence, with a unity absorptance ranging from 3.5 THz to 5 THz range. The simple design, featuring a single elliptical ring, minimizes fabrication complexity, while the phase transition of VO₂ enables dynamic tuning of the absorption peak from 0.05% to 100% via temperature adjustment, allowing the model to also function as a near-zero absorption perfect reflector when VO₂ is in its insulating state (200 S/m). The absorption mechanism is validated through wave-interference and impedance matching theories, supported by electric field distribution analysis at key frequencies. Geometric optimization maximizes bandwidth and efficiency, and eleven machine learning techniques—enhanced by data augmentation and feature engineering—were employed to predict optimal parameters, with the Random Forest Regressor performing best for most targets and the CATBoost Regressor excelling in the remaining cases, yielding predictions closely aligned with simulations. Additionally, the model demonstrates wide-angle absorption, maintaining over 80% absorptance

in TE mode up to 70° , making it highly suitable for THz applications like modulators, optical switches, sensors, and other optical devices.

REFERENCES

- [1] R. I. Stantchev, X. Yu, T. Blu, and E. Pickwell-MacPherson, "Real-time terahertz imaging with a single-pixel detector," *Nature Commun.*, vol. 11, no. 1, p. 2535, 2020.
- [2] J. Ma, R. Shrestha, J. Adelberg, C.-Y. Yeh, Z. Hossain, E. Knightly, J. M. Jornet, and D. M. Mittleman, "Security and eavesdropping in terahertz wireless links," *Nature*, vol. 563, no. 7729, pp. 89–93, 2018.
- [3] S. Koenig, D. Lopez-Diaz, J. Antes, F. Boes, R. Henneberger, A. Leuther, A. Tessmann, R. Schmogrow, D. Hillerkuss, and R. Palmer, "Wireless sub-THz communication system with high data rate," *Nature Photon.*, vol. 7, no. 12, pp. 977–981, Dec. 2013.
- [4] X. Lu, F. Zhang, L. Zhu, S. Peng, J. Yan, Q. Shi, K. Chen, X. Chang, H. Zhu, C. Zhang, W. Huang, and Q. Cheng, "A terahertz meta-sensor array for 2D strain mapping," *Nature Commun.*, vol. 15, no. 1, p. 3157, Apr. 2024.
- [5] D.-S. Yang, T. Jiang, and X.-A. Cheng, "Optically controlled terahertz modulator by liquid-exfoliated multilayer WS₂ nanosheets," *Opt. Exp.*, vol. 25, no. 14, pp. 16364–16377, Jul. 2017.
- [6] K.-T. Lin, H. Lin, T. Yang, and B. Jia, "Structured graphene metamaterial selective absorbers for high efficiency and omnidirectional solar thermal energy conversion," *Nature Commun.*, vol. 11, no. 1, p. 1389, Mar. 2020.
- [7] N. I. Landy, S. Sajuyigbe, J. J. Mock, D. R. Smith, and W. J. Padilla, "Perfect metamaterial absorber," *Phys. Rev. Lett.*, vol. 100, no. 20, 2008, Art. no. 207402.
- [8] J. Huang, J. Li, Y. Yang, J. Li, J. Li, Y. Zhang, and J. Yao, "Active controllable dual broadband terahertz absorber based on hybrid metamaterials with vanadium dioxide," *Opt. Exp.*, vol. 28, no. 5, pp. 7018–7027, Mar. 2020.
- [9] J. Cheng, H. Zhang, M. Ning, H. Raza, D. Zhang, G. Zheng, Q. Zheng, and R. Che, "Emerging materials and designs for low- and multi-band electromagnetic wave absorbers: The search for dielectric and magnetic synergy?" *Adv. Funct. Mater.*, vol. 32, no. 23, Jun. 2022, Art. no. 2200123.
- [10] G. Wu, C. Li, D. Wang, S. Gao, W. Chen, S. Guo, and J. Xiong, "Vanadium dioxide-based ultra-broadband metamaterial absorber for terahertz waves," *Opt. Mater.*, vol. 147, Jan. 2024, Art. no. 114667.
- [11] P. Zhang, K. Qin, X.-H. Deng, H. Liu, and J. Yuan, "Tunable ultra-wideband polarization insensitivity wide-angle perfect THz absorber based on metamaterials containing vanadium dioxide," *Appl. Phys. A, Solids Surf.*, vol. 129, no. 9, p. 644, Sep. 2023.
- [12] Y. Zou, H. Lin, Y. Wu, H. Zhu, X. Zhang, and B.-X. Wang, "Theoretical investigation of an ultra-wideband tunable metamaterial absorber based on four identical vanadium dioxide resonators in the terahertz band," *J. Electron. Mater.*, vol. 52, no. 4, pp. 2852–2864, Apr. 2023.
- [13] H. Meng, Z. Cui, P. Wu, S. Zhang, and L. Wang, "Superior carrier mobility and photogalvanic effect of Si₃C₁₅ and WSSe heterojunctions," *Mater. Today Commun.*, vol. 46, Jun. 2025, Art. no. 112830.
- [14] D. Shrekenhamer, W.-C. Chen, and W. J. Padilla, "Liquid crystal tunable metamaterial absorber," *Phys. Rev. Lett.*, vol. 110, no. 17, Apr. 2013, Art. no. 177403.
- [15] S. J. Kim, P. Fan, J.-H. Kang, and M. L. Brongersma, "Creating semiconductor metafilms with designer absorption spectra," *Nature Commun.*, vol. 6, no. 1, p. 7591, Jul. 2015.
- [16] G. Wu, X. Jiao, Y. Wang, Z. Zhao, Y. Wang, and J. Liu, "Ultra-wideband tunable metamaterial perfect absorber based on vanadium dioxide," *Opt. Exp.*, vol. 29, no. 2, pp. 2703–2711, Jan. 2021.
- [17] J. King, C. Wan, T. J. Park, S. Deshpande, Z. Zhang, S. Ramanathan, and M. A. Kats, "Electrically tunable VO₂–metal metasurface for mid-infrared switching, limiting and nonlinear isolation," *Nature Photon.*, vol. 18, no. 1, pp. 74–80, Jan. 2024.
- [18] G. Li, Z. Wang, Y. Chen, J.-C. Jeon, and S. S. P. Parkin, "Computational elements based on coupled VO₂ oscillators via tunable thermal triggering," *Nature Commun.*, vol. 15, no. 1, p. 5820, Jul. 2024.
- [19] X. Xu, R. Xu, and Y.-S. Lin, "A voltage-controllable VO₂ based metamaterial perfect absorber for CO₂ gas sensing application," *Nanoscale*, vol. 14, no. 7, pp. 2722–2728, Feb. 2022.
- [20] Z.-M. Liu, Y. Li, J. Zhang, Y.-Q. Huang, Z.-P. Li, J.-H. Pei, B.-Y. Fang, X.-H. Wang, and H. Xiao, "Design and fabrication of a tunable infrared metamaterial absorber based on VO₂ films," *J. Phys. D, Appl. Phys.*, vol. 50, no. 38, Sep. 2017, Art. no. 385104.
- [21] M. Zhong, "Design and verification of a temperature-sensitive broadband metamaterial absorber based on VO₂ film," *Opt. Mater.*, vol. 109, Nov. 2020, Art. no. 110467.
- [22] Y. Zhu, "Vanadium dioxide-based tunable broadband metamaterial absorber in the terahertz," *Mater. Res. Exp.*, vol. 12, no. 1, Jan. 2025, Art. no. 015801.
- [23] S. Wang, C. Cai, M. You, F. Liu, M. Wu, S. Li, H. Bao, L. Kang, and D. H. Werner, "Vanadium dioxide based broadband THz metamaterial absorbers with high tunability: Simulation study," *Opt. Exp.*, vol. 27, no. 14, p. 19436, Jul. 2019.
- [24] Z. Song, M. Wei, Z. Wang, G. Cai, Y. Liu, and Y. Zhou, "Terahertz absorber with reconfigurable bandwidth based on isotropic vanadium dioxide metasurfaces," *IEEE Photon. J.*, vol. 11, no. 2, pp. 1–7, Apr. 2019.
- [25] P. Zhang, G. Chen, Z. Hou, Y. Zhang, J. Shen, C. Li, M. Zhao, Z. Gao, Z. Li, and T. Tang, "Ultra-broadband tunable terahertz metamaterial absorber based on double-layer vanadium dioxide square ring arrays," *Micromachines*, vol. 13, no. 5, p. 669, Apr. 2022.
- [26] W. Zhao, J. Wang, R. Li, and B. Zhang, "Ultrabroad dual-band metamaterial perfect absorber and its sensing application," *J. Opt.*, vol. 24, no. 3, Mar. 2022, Art. no. 035103.
- [27] A. S. Karim, "Simulation and experimental studies of broadband multi-resonators metamaterial absorber for satellite communications," *Bull. Mater. Sci.*, vol. 47, no. 2, p. 114, Jun. 2024.
- [28] H. Zhang, X.-Y. Cao, J. Gao, H. Yang, and Q. Yang, "A novel dual-band metamaterial absorber and its application for microstrip antenna," *Prog. Electromagn. Res. Lett.*, vol. 44, pp. 35–41, 2014.
- [29] P. Rufangura and C. Sabah, "Dual-band perfect metamaterial absorber for solar cell applications," *Vacuum*, vol. 120, pp. 68–74, Oct. 2015.
- [30] Z. Cui, Z. Diao, X. Wang, X. Gao, S. Zhang, and L. Wang, "AlN/Ti₂CO₂ heterojunction photodetector with high carrier mobility and photore-sponse," *Micro Nanostruct.*, vol. 207, Nov. 2025, Art. no. 208268.
- [31] Z. Cui, H. Wang, Y. Luo, E. Li, Y. Shen, K. Qin, and P. Yuan, "Mxenes/WSSe heterojunction photodetector with ultrahigh sensitivity and accuracy," *Appl. Surf. Sci.*, vol. 684, Mar. 2025, Art. no. 161853.
- [32] Z. Cui, H. Meng, C. Zhang, L. Zhang, S. Zhang, and L. Wang, "MoSSe/Si₃C₁₅ heterojunction photodetectors with ultrahigh photocurrent and carrier mobility," *Mater. Sci. Semicond. Process.*, vol. 182, Nov. 2024, Art. no. 108705.
- [33] Y. Zhu, Z. Cui, S. Zhang, and L. Wang, "Dual-frequency/broadband switchable and dynamically tunable metamaterial terahertz absorber based on graphene and VO₂," *Solid State Commun.*, vol. 404, Oct. 2025, Art. no. 116087.
- [34] X. Zhao, Y. Shen, Z. Cui, K. Qin, D. Ma, F. Cheng, P. Yuan, X. Qi, and E. Li, "Performance modulation of ZnO/WSSe heterojunction photodetectors via longitudinal stress bending," *Chin. J. Phys.*, vol. 98, pp. 137–146, Dec. 2025.
- [35] M. Jiang, Y. Li, and H. Zhang, "Corrosion resistance and plasma surface treatment on titanium and titanium alloys: A review," *Coatings*, vol. 15, no. 10, p. 1180, Oct. 2025.
- [36] P. Koukouvini and J. Anagnostopoulos, "State of the art in designing fish-friendly turbines: Concepts and performance indicators," *Energies*, vol. 16, no. 6, p. 2661, Mar. 2023.
- [37] S. K. Patel, J. Parmar, and V. Katkar, "Ultra-broadband, wide-angle plus-shape slotted metamaterial solar absorber design with absorption forecasting using machine learning," *Sci. Rep.*, vol. 12, no. 1, p. 10166, Jun. 2022.
- [38] P. Jain, H. Chhabra, U. Chauhan, K. Prakash, A. Gupta, M. S. Soliman, M. S. Islam, and M. T. Islam, "Machine learning assisted hepta band THz metamaterial absorber for biomedical applications," *Sci. Rep.*, vol. 13, no. 1, p. 1792, Jan. 2023.
- [39] B. S. Chouhan, A. Nawaz, A. Das, K. M. Rohith, A. Ahmad, and G. Kumar, "Machine learning-driven ultra-broadband terahertz multilayer metamaterial," *J. Lightw. Technol.*, vol. 43, no. 6, pp. 2734–2744, Mar. 15, 2025.
- [40] M. Soni and S. Misra, "Machine-learning-assisted design of multiband terahertz metamaterial absorber," *ACS Appl. Opt. Mater.*, vol. 1, no. 10, pp. 1679–1687, Oct. 2023.
- [41] Z.-X. Liu, J. Jin, L.-J. Chen, J.-P. Fu, and H. Lin, "Metamaterial absorber optimization method based on an artificial neural network surrogate," *Opt. Exp.*, vol. 31, no. 22, p. 35594, Oct. 2023.

- [42] B.-X. Wang, C. Xu, G. Duan, W. Xu, and F. Pi, "Review of broadband metamaterial absorbers: From principles, design strategies, and tunable properties to functional applications," *Adv. Funct. Mater.*, vol. 33, no. 14, Apr. 2023, Art. no. 2213818.
- [43] S. Liu, X. Yin, and H. Zhao, "The tunable single-/narrow-band terahertz metamaterial absorber through photoconductivity," *Results Phys.*, vol. 39, Aug. 2022, Art. no. 105741.
- [44] H. Ge, K. Jia, Y. Jiang, Y. Bu, Y. Zhang, Y. Zhang, and Q. Sun, "Dual-function terahertz metamaterial absorber based on microfluidic structures," *Opt. Exp.*, vol. 33, no. 3, p. 3930, Feb. 2025.
- [45] K.-J. Ri, R.-J. Kang, and C.-H. Ri, "Tunable ultra-broadband terahertz metamaterial absorbers based on complementary split ring-shaped graphene," *AIP Adv.*, vol. 14, no. 5, May 2024, Art. no. 055301.
- [46] Y. Wu, W. Xu, H. Zhou, X. Qiu, Y. He, Y. Gao, and B.-X. Wang, "Tunableness of single-band and dual-band absorption and filtering using vanadium-dioxide-based metamaterial," *Appl. Phys. A, Solids Surf.*, vol. 128, no. 10, p. 930, Oct. 2022.
- [47] Y. Wang, Y. Chen, F. Liu, L. Chen, K. Ji, X. Wang, and X. Ji, "Vanadium dioxide enabled polarization insensitive tunable broadband terahertz metamaterial absorber," *Sci. Rep.*, vol. 15, no. 1, p. 10140, Mar. 2025.
- [48] P. Bhatt, P. Jain, and A. Joshi, "Machine learning-assisted multi-band metamaterial absorber for terahertz sensing and gas detection," *Phys. Scripta*, vol. 100, no. 7, Jul. 2025, Art. no. 075530.
- [49] P. Jain, M. T. Islam, and A. S. Alshammari, "Comparative analysis of machine learning techniques for metamaterial absorber performance in terahertz applications," *Alexandria Eng. J.*, vol. 103, pp. 51–59, Sep. 2024.
- [50] P. Jain, H. Chhabra, U. Chauhan, K. Prakash, P. Samant, D. K. Singh, M. S. Soliman, and M. T. Islam, "Machine learning techniques for predicting metamaterial microwave absorption performance: A comparison," *IEEE Access*, vol. 11, pp. 128774–128783, 2023.
- [51] C. Ba, S. T. Bah, M. D'Auteuil, P. V. Ashrit, and R. Vallée, "Fabrication of high-quality VO₂ thin films by ion-assisted dual AC magnetron sputtering," *ACS Appl. Mater. Interfaces*, vol. 5, no. 23, pp. 12520–12525, Dec. 2013.
- [52] L. Xiao, H. Ma, J. Liu, W. Zhao, Y. Jia, Q. Zhao, K. Liu, Y. Wu, Y. Wei, S. Fan, and K. Jiang, "Fast adaptive thermal camouflage based on flexible VO₂/graphene/CNT thin films," *Nano Lett.*, vol. 15, no. 12, pp. 8365–8370, Dec. 2015.
- [53] E. K. Barimah, A. Boontan, D. P. Steenson, and G. Jose, "Infrared optical properties modulation of VO₂ thin film fabricated by ultrafast pulsed laser deposition for thermochromic smart window applications," *Sci. Rep.*, vol. 12, no. 1, p. 11421, Jul. 2022.
- [54] T. Yamin, T. Havdala, and A. Sharoni, "Patterning of epitaxial VO₂ microstructures by a high-temperature lift-off process," *Mater. Res. Exp.*, vol. 1, no. 4, Oct. 2014, Art. no. 046302.
- [55] N. D. Bassim, A. J. Giles, L. E. Ocola, and J. D. Caldwell, "Fabrication of phonon-based metamaterial structures using focused ion beam patterning," *Appl. Phys. Lett.*, vol. 112, no. 9, Feb. 2018, Art. no. 091101.
- [56] S. Kim, Y. Xuan, V. P. Drachev, L. T. Varghese, L. Fan, M. Qi, and K. J. Webb, "Nanoimprinted plasmonic nanocavity arrays," *Opt. Exp.*, vol. 21, no. 13, p. 15081, Jul. 2013.



application, contributing to advancements in both fundamental engineering principles and modern computational methods.

OISHI JYOTI (Member, IEEE) received the B.Sc. degree in electrical and computer engineering and the M.Sc. degree in electrical and electronic engineering from Rajshahi University of Engineering and Technology, Bangladesh. With a keen interest in cutting-edge technologies, her research focuses on the innovative field of metamaterials, alongside the integration of machine learning techniques in simulation and inverse design. Her work aims to bridge the gap between theory and practical



of electrical and electronic engineering with RUET. His current research interests include the applications of metamaterials and metasurfaces in nanophotonics, increasingly leveraging machine learning techniques to innovate photonic device design and functionality. He is a member of Optica and a fellow of the Institution of Engineers, Bangladesh (IEB). During his Ph.D. degree, he received the prestigious Australian Government International Postgraduate Research Scholarship (IPRS).



NGUYEN HOANG HAI (Member, IEEE) was born in Hanoi, Vietnam, in 1977. He received the B.E. degree in electronics and telecommunication engineering from Hanoi University of Technology (HUT), in 1999, and the Ph.D. degree in electrical and electronics engineering from the University of the Ryukyus, Okinawa, Japan, in 2009. In October 2000, he joined the Department of Electrical and Electronics Engineering, University of the Ryukyus, as a Graduate Student, where he worked in the field of microwave circuit elements and electromagnetic theory. After completion of his Ph.D. degree, he joined the Department of Telecommunication System, School of Electronics and Telecommunication, Hanoi University of Science and Technology, in 2009. In 2013, he became an Associate Professor with the Department of Telecommunication System, School of Electronics and Telecommunication, HUST, where he became the Deputy Head of the Department of Telecommunication System, in 2013. He has been engaged in research on analysis of optical fiber transmission characteristics, chromatic dispersion (CD), polarization characteristics, polarization mode dispersion (PMD), and highly nonlinear effect in optical systems, in single mode optical fibers and optical fiber submarine systems. Recently, he has focused his research on optimized designs for various types of photonic crystal fibers (PCFs) and its application for broadband transmission systems, and medical application as well.



S. M. ABDUR RAZZAK (Senior Member, IEEE) was born in Natore, Bangladesh, in December 1974. He received the B.Sc. degree from Rajshahi University of Engineering and Technology (formerly B.I.T. Rajshahi), Rajshahi, Bangladesh, in 1998, and the M.Eng. and Ph.D. degrees in electrical and electronic engineering from the University of the Ryukyus, Okinawa, Japan, in 2007 and 2010, respectively. He is a fellow of the Institution of Engineers, Bangladesh, and a Student Member of the Institute of Electronics, Information and Communication Engineers, Japan, and the Optical Society of America. He received the President's Honorary Award from the University of the Ryukyus, in 2007, and the Board of Governors' Gold Medal from B.I.T. Rajshahi, in 2000.

...